Prem **Mallappa**

Prem Mallappa - CV (2-page) Software Architect • System & Embedded Software

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Seasoned Software Architect with extensive expertise in designing and developing high-performance embedded systems, system software, Linux kernel drivers, and virtualization technologies. Proficient in a wide range of programming and scripting languages, with a proven track record of delivering robust, scalable solutions in complex technical environments.

A highly analytical and collaborative team player, recognized for strong problem-solving abilities, logical thinking, and a passion for mastering emerging technologies. Committed to driving innovation and excellence through clean architecture, optimized code, and best-in-class engineering practices.

Skills

Leadership	Over 22 years of multidisciplinary team and project leadership experience; Software Architecture and Design.
Communications	Excellent written and spoken communication skills.
Technology	Broad and deep IT expertise, including cloud computing, computer security, operating systems, embedded systems, software & services development, programming languages, etc.
Computer Security	Cryptography; virtualization and cloud computing security architecture and implementation; risk management and compliance; intrusion detection and prevention; software security and secure software development.
Programming Languages	C, Assembly (x86, ARM, MIPS), Rust, Python, C++, Go, BASH
Operating Systems	Linux, FreeBSD, QNX, VxWorks, Symbian
Architecture	x86, ARM - v6,v7,v8, MIPS64, PowerPC, RISC-V
Development Tools	GCC, Clang, Git, Make, CMake, ARM Development Tools, Keil, JTAG, Docker,
Specialized Skills	Linux Kernel Development, Device Drivers, Embedded Systems, System Software, Performance Optimization, Cryptography, Filesystem Development

Education

Masters (Computer Science)
Pilani, Rajasthan, India

BITS Pilani
2016

Bachelors (Computer Science)
Chikkmagalur, Karnataka, India

Adichunchangiri Institute of Technology (AIT)
2002

Affiliated: Visvesvaraya Technological University (VTU)

Open Source

QEMU (Link)
2014-2016

- SMMUv3 (IOMMU) emulation support for ARMv8
- Designed and implemented SMMUv3 model merged into mainline
- Added support for Stage1, Stage2, and nested virtualization
- Implemented command queue processing and page table walk

Linux Kernel (Link)
2011-2016

- MIPS Kexec/Kdump port and IOMMU subsystem
- Developed MIPS64 port of Kexec and Kdump (merged upstream)

- Fixed critical bugs in KEXEC for Cavium Octeon platforms
- Contributed to IOMMU/SMMUv3 driver development

GLIBC (Link)

2018-2020

- Performance optimizations for AMD processors
- Fixed memcpy behaviour on AMD processors
- Optimized string functions for x86\\$_{64}$ architecture
- Performance improvements for memory operations

AMD LibM (Link)

2018-2025

- Open source math library for AMD processors
- Core contributor to open sourcing AMD Math Library
- Optimized transcendental functions (exp, log, pow, trig)
- Implemented SIMD/FMA optimizations for vector operations

Experience

AMD Ltd.

2018 – Presen

Bengaluru, Ind

PRINCIPAL ENGINEER

Worked in multiple teams within AMD including **AMD LibM**, **AOCL Cryptography**, **BSP**, and **NVMe** in **Pensando** BU. Every project was different, requiring quick learning and adaptation to new domains. The role also involved leading a team of four-to-six engineers in various projects. I was also part of the leadership team where I mentored engineers from across AOCL and Pensando BU.

SR. MEMBER OF TECHNICAL STAFF

Worked on Model0 project which was pre-silicon validation of next generation AMD cores, including Zen2 and Zen3. The role involved working with hardware team to bring up bare-minimum processor configuration with just core and memory controller, and run Linux, stressors and benchmarks. This involved working on low-level firmware, bootloaders, Linux kernel and performance analysis tools.

Broadcom Ltd.

2014 – 2016

Bengaluru, Ind

PRINCIPAL ENGINEER

Architected foundational software stack for Broadcom Vulcan, a multicore-multithreaded ARMv8 64-bit server processor. Engineered complete SMMUv3 (System Memory Management Unit) driver infrastructure supporting advanced IOMMU virtualization. Developed and contributed SMMUv3 emulation model to QEMU mainline, enabling early platform validation and ecosystem enablement. Implemented comprehensive virtualization features including command queue processing, STE/CD descriptor parsing, multi-level page table walks, and stage-1/stage-2 address translation.

Cavium India Pvt. Ltd.

2011 – 2013

Bengaluru, Ind

TECH LEAD

Led low-level software development for Cavium Octeon III (MIPS64) multicore network processors, delivering stable platform firmware for networking customers. Resolved a critical KEXEC kernel bug and authored the full MIPS64 port of the Kexec/Kdump crash dump infrastructure, ultimately merged into Linux mainline. Engineered a bare-metal core dump generation framework with a host-resident daemon that enabled detailed post-mortem GDB analysis during bring-up. Designed and implemented the CavHv hypervisor for MIPS64 architecture using experimental hardware virtualization extensions to validate next-generation features.

ARM Ltd.

2005 – 2009

Bengaluru, Ind

SR. DEVELOPMENT ENGINEER

Executed comprehensive OS porting initiatives for emerging ARM architectures including ARM1176JZFS with TrustZone security extensions and the Cortex-A8 application processor. Developed a production-grade touchscreen driver for Symbian OS along with a Python-based automated validation framework used across multiple programs. Designed a precision interrupt latency measurement infrastructure that quantified TrustZone secure-world context switching overhead for silicon partners. Ported the L4Ka::Pistachio microkernel to ARM architecture, enabling virtualization research and multiple proof-of-concept implementations.

Sasken Communication Technologies Ltd.

2004 – 2005

Bengaluru, Ind

SR. SOFTWARE ENGINEER

Architected and maintained the Extended File System (EFS) for VxWorks-based UMTS base station infrastructure, balancing throughput with resiliency requirements. Engineered a reset-resilient filesystem with advanced wear-leveling algorithms so data integrity held across unexpected power cycles in the field. Designed a flash-optimized flat file tree structure that minimized write amplification and significantly extended device longevity.

Global Edge Software Ltd.

2003 – 2004

Bengaluru, Ind

SOFTWARE ENGINEER

Designed and implemented a comprehensive SDIO host controller driver for Linux on Intel StrongARM embedded platforms used in consumer devices. Engineered a high-performance 4-bit SDIO driver for Marvell 802.11g WiFi chipsets that consistently achieved maximum theoretical throughput. Optimized DMA transfers and interrupt handling paths to minimize latency and maximize wireless data transfer efficiency under heavy load.

Short Stints

VSPL Ltd.

Bengaluru, India

Software Architect

Nov. 2016 – Jan. 2018

Cisco Ltd.

Bengaluru, India

Software Engineer

Oct. 2010 – Apr. 2011

B-Labs, London UK

Bengaluru, India

Sr. Engineer - Contractor

Nov. 2009 – Sep. 2010

Harman International

Bengaluru, India

Engineer

Jul. 2009 – Nov. 2009